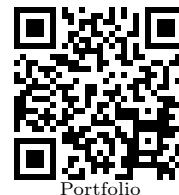


Aditya Khadye

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Portfolio

Professional Summary

Data Scientist & ML Engineer focused on production-scale pipelines and Transformer architectures. Optimized Snowflake ETL runtimes by **39% (\$200K/yr savings)** at Travel + Leisure Co. Currently leading the **ScriptGap** research project at UCF, developing cross-lingual embedding frameworks sponsored by **Google, Adobe, and Monotype**. Specialist in high-throughput GPU training, CI/CD automation, and engineering real-time data ingestion for XR platforms.

Education

University of Central Florida

Expected Aug 2026

M.S. in Data Analytics (Machine Learning Track) — Graduate Research Assistant (CECS/IE)

Course GPA 3.8 — Focused on Transformer-based Architectures, NLP, and Distributed Data Systems.

University of North Florida

Dec 2024

B.S. in Computer Science — Honors in Computing; Dean's List (5x)

Course GPA 3.9 — IEEE 17.0 Hackathon **Top 5% Global Placement** (800+ teams).

Professional Experience

Graduate Research Assistant (Lead Researcher)

Jun 2025 – Present

University of Central Florida — TRC / MIT² Lab, Orlando, FL

- Engineering **Transformer-based embedding pipelines** for the ScriptGap project, analyzing global digital reading signals and processing font metadata to guide strategic R&D for **Google, Adobe, and Monotype**.
- Architected real-time **multi-stream data ingestion pipelines** for Apple Vision Pro, utilizing TCP streaming to convert raw sensor logs into analysis-ready datasets for high-fidelity ML experimentation.
- Deployed and debugged distributed workloads on **Linux clusters (Slurm + MPI)**, improving workload repeatability and reducing experiment reruns by 40% through automated validation gates and runbooks.

Data Science Intern

May 2024 – Aug 2024

Travel + Leisure Co., Orlando, FL

- **Reduced Snowflake ETL runtime by 39% and generated \$200K/yr in savings** via advanced SQL tuning, query optimization, and warehouse right-sizing for high-scale enterprise production environments.
- Built and productionized tree-based models to forecast pipeline failures; deployed on **AWS with Redis** for low-latency serving, enabling proactive data quality intervention and reducing failed jobs by **28%**.
- Established robust data governance and monitoring standards including lineage, SLAs, and automated recovery procedures; improved overall data quality to **99%** across mission-critical ML workflows.

Lead Computer Science Tutor

Aug 2023 – May 2024

University of North Florida, Jacksonville, FL

- Mentored **500+ students** in C++, Data Structures & Algorithms, and Debugging; focused on teaching **clean code principles, unit testing, and defect-free software development** practices.

Key Projects

ScriptGap: Global Reading Analytics & Transformer Framework

Jan 2026 – Present

PyTorch, Transformers, LLMs, NLP, GCP, BigQuery, OpenCV, Latent Space Analysis

- Developing an end-to-end ML framework to rank global writing systems by typographic support using **Latent Space Analysis** of font datasets (**OTF**) and web reading signals (**CRUX**).
- Delivering actionable, data-driven reports and visualizations to identify underserved languages, guiding prioritization decisions for major industry partners including **Google and Adobe**.

ViT (Vision Transformer): Training Performance Optimization

Apr 2025

PyTorch, Transformers, GPU Training, Mixed Precision, Input Pipelines

- Increased training throughput to **1,240 images/sec** and boosted GPU utilization to **92%** by implementing mixed-precision training and input pipeline tuning, reducing total training time by **36%**.

Agentic AI RAG System with Guardrails

Dec 2024

Transformers, FAISS, FastAPI, AWS, Redis, Gemini/OpenAI APIs, Hybrid Retrieval

- Built a knowledge-base RAG pipeline utilizing hybrid retrieval (**BM25 + Vector Search**), cached summaries, and quantized inference for low-latency summarized Q&A with built-in evaluation guardrails.

Technical Skills

Languages: Python (Expert), SQL (PostgreSQL/Snowflake), C++, Java, R

AI/ML/NLP: Generative AI, LLMs, Transformers, NLP, PyTorch, Hugging Face, Scikit-learn, Mixed Precision Training

MLOps/DevOps: CI/CD (Jenkins, GitHub Actions, ArgoCD), Infrastructure as Code (Terraform), Docker, Kubernetes, Slurm

Data Engineering: Snowflake (Optimization/Modeling), ETL/ELT, AWS (S3, EC2, IAM), Redis, Spark, BigQuery, NoSQL

Systems/Tools: Linux, Git, REST APIs, FastAPI, Agile/Scrum, Distributed Computing (MPI)

Certifications

- IBM — Continuous Integration and Continuous Delivery (2026)
- Amazon — Machine Learning on AWS (2024)
- Microsoft — Azure SQL (2026)
- Stanford — Advanced Learning Algorithms (2024)
- Google — Agile & Scrum Project Mgmt (2025)
- IBM — Machine Learning with Python (2024)